

MSc Project Management Tools: Agile Methodology

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In the realm of computing project management, the strategic utilization of Kanban Boards, Flowcharts, and Gantt Charts proves indispensable for ensuring efficiency, clarity, and successful project outcomes. Kanban Boards facilitate visualizing the workflow, providing a real-time snapshot of tasks and their status. This visual representation enhances collaboration and enables the team to identify and address bottlenecks swiftly, crucial in the dynamic landscape of computing development. Flowcharts serve as an invaluable tool for mapping out intricate processes and dependencies within MSc projects. Their visual nature aids in comprehending complex algorithms and decision trees, fostering better communication among team members. Gantt Charts, with their timeline-based structure, offer a comprehensive overview of project timelines, task dependencies, and resource allocation. This aids in effective project scheduling, enabling teams to manage MSc projects with precision and meet milestones seamlessly. Together, these tools create a robust project management framework that is essential for navigating the complexities inherent in computing development.

Kanban Boards

In this course, we'll be using a Kanban board with three columns: 'To Do,' 'Doing,' and 'Done' (Figure 1). The Kanban board is a visual tool that helps us manage tasks and projects effectively as a team. When we start working on a project or assignment, we place it in the 'To Do' column. However, before we begin, we'll break down the problem into individual tasks. As a group, you'll identify all the tasks required to solve the problem or complete the project and list them on your Kanban board. Each task will have a clear description, outlining what needs to be done.

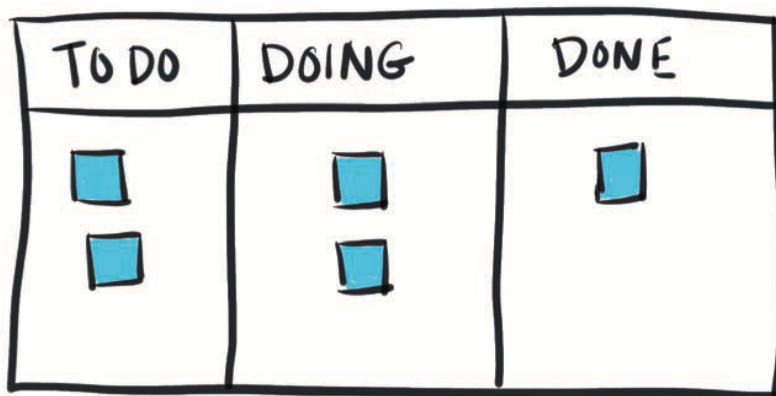


Figure 1. A simple three column Kanban board. There are more complex ones that incorporate into different kinds of projects. The image was taken from the book: DeGrandis, D. (2022). *Making work visible: Exposing time theft to optimize work & flow*. IT Revolution.

Assigning responsibilities: We'll consider everyone's strengths, skills, and preferences to ensure that each task is tackled by the right team member. Some tasks may have a single responsible person, while larger tasks may involve collaboration among team members. Once we've assigned tasks, we'll move them from 'To Do' to 'Doing' on our Kanban board, which will serve as our visual project management tool.

Make your work visible: Throughout the seminar, collaborate closely and have regular check-ins to discuss progress, challenges, and any necessary adjustments. As you make progress, update the status of our tasks on the Kanban board, moving them from 'Doing' to 'Done' upon

completion. This approach not only helps us work efficiently but also fosters teamwork and transparency. So, get ready to break down problems, assign tasks, and visualize our progress on our Kanban boards as we embark on exciting group projects!

Gantt Charts

A Gantt chart is a visual project management tool that displays tasks or activities along a timeline. It's named after its creator, Henry L. Gantt. Gantt charts provide a clear and structured way to plan, schedule, and track tasks within a project. Each task is represented as a bar on the chart, with the length of the bar indicating the task's duration. Gantt charts allow teams to see task dependencies, set milestones, allocate resources, and monitor project progress. They are especially valuable for complex projects with multiple team members and various activities. In a group projects, teams can create Gantt charts to outline their project timelines, allocate responsibilities, and ensure tasks are completed on time. It's a powerful tool for visualizing project schedules and ensuring that everyone is on the same page regarding project timelines and deadlines.

For this course, we will use a Gantt Chart that looks like this:

TASK NAME	0-5 mins	5-10 mins	10-15 mins	15-20 mins	20-25 mins	25-30 mins
Task 1 Understanding the Problem	Scrum Master Makes sure problem is understood					
Task 2		Student 1 Student 2 Work on task 2	Student 1 Student 2 Review solutions	Scrum Master reviews progress		
Task 3			Student 3 Student 4 Work on task 2	Student 1 Student 2 Review solutions	Scrum Master reviews progress	
Task 4 Presenting						Presenter

Table 1. A Gantt chart is a visual representation of a project schedule that uses horizontal bars to illustrate the timing and duration of various tasks or activities. It typically includes a timeline along the horizontal axis and a list of tasks or activities along the vertical axis. Each task is represented as a separate bar on the chart, positioned along the timeline based on its start and end dates.

Flow Charts

A flowchart is a visual representation of a process or a series of steps. It consists of various shapes and symbols connected by arrows or lines, each representing a specific action, decision, or task in a logical sequence. Key elements are:

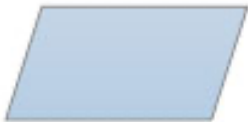
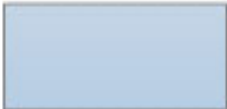
Symbol	Name	Function
	Start/end	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/Output	A parallelogram represents input or output
	Process	A rectangle represents a process
	Decision	A diamond indicates a decision

Figure 2. Typical shapes for flowchart creation. The shape of the element tells programmers what kind of processes are involved at a first glance.

1. **Start and End Symbols:** A flowchart typically begins with a "Start" symbol, which indicates where the process starts, and ends with an "End" symbol, marking the conclusion of the process.
2. **Flow Arrows:** Flow arrows, connecting the various symbols, indicate the direction of the process. These arrows show the sequence of actions, decisions, or events.
3. **Input/Output Symbols:** Parallelograms are used to depict input or output steps. They signify where data is entered into the process or where results are produced.
4. **Process Symbols:** Rectangles are used to represent process steps or actions. These boxes describe what action is taken at that stage of the process.
5. **Decision Symbols:** Diamond shapes are used to represent decision points or conditional statements. Depending on the result of the decision, the flow can take different paths, often labelled "Yes" and "No."

Flowcharts are used to break down problems into smaller, manageable steps, making it easier to identify solutions. Furthermore, they are an effective way to document processes, procedures, and a universal language for conveying process-related information.